

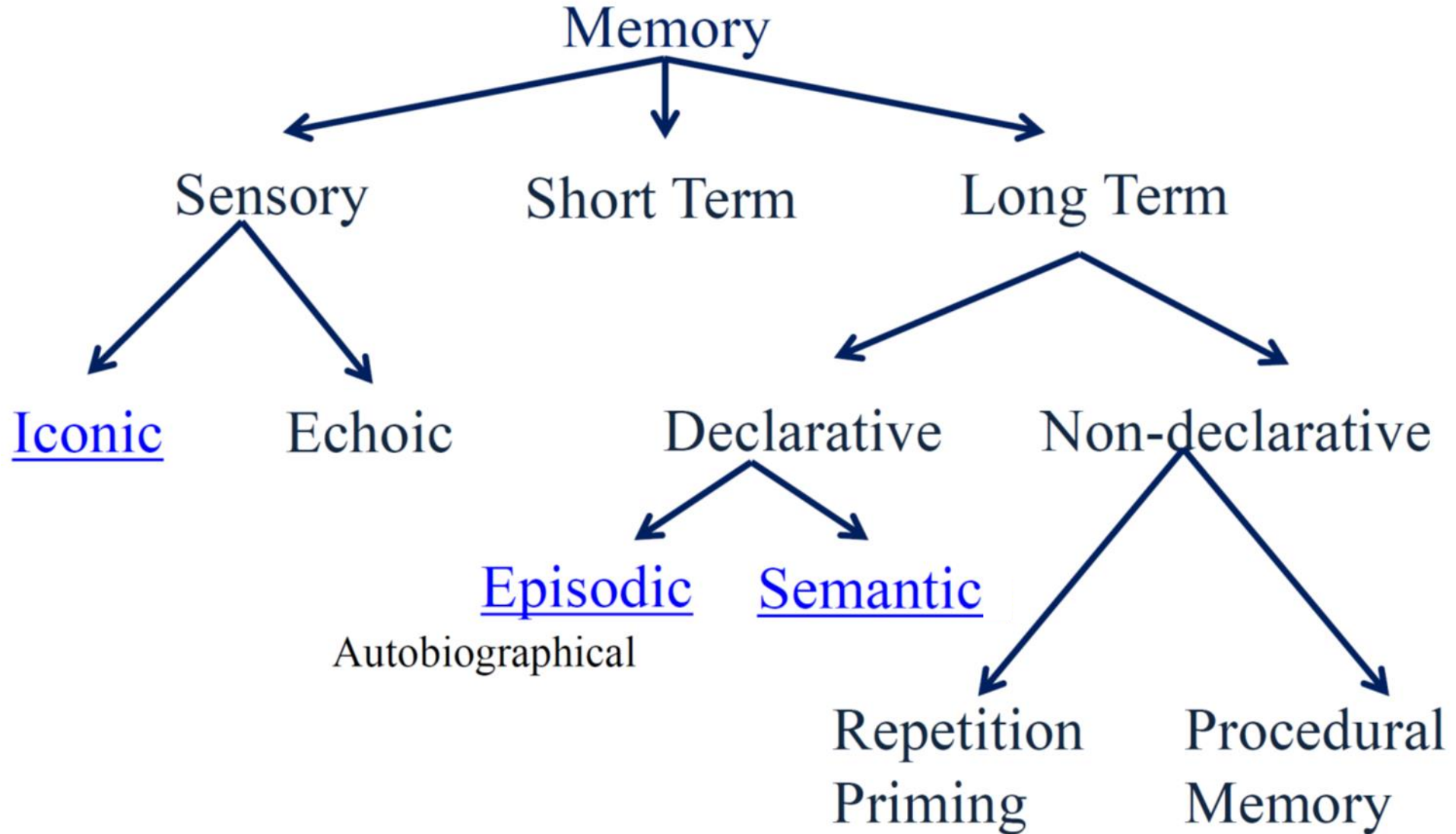


Motor System System - Part 1

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Memory Taxonomy



Procedural Memory - Special?

- Unlike declarative memory (semantic memory of facts)
- Takes long time to establish
- Hard to erase (forget)
 - Ex: Skill memory: riding a bicycle, etc

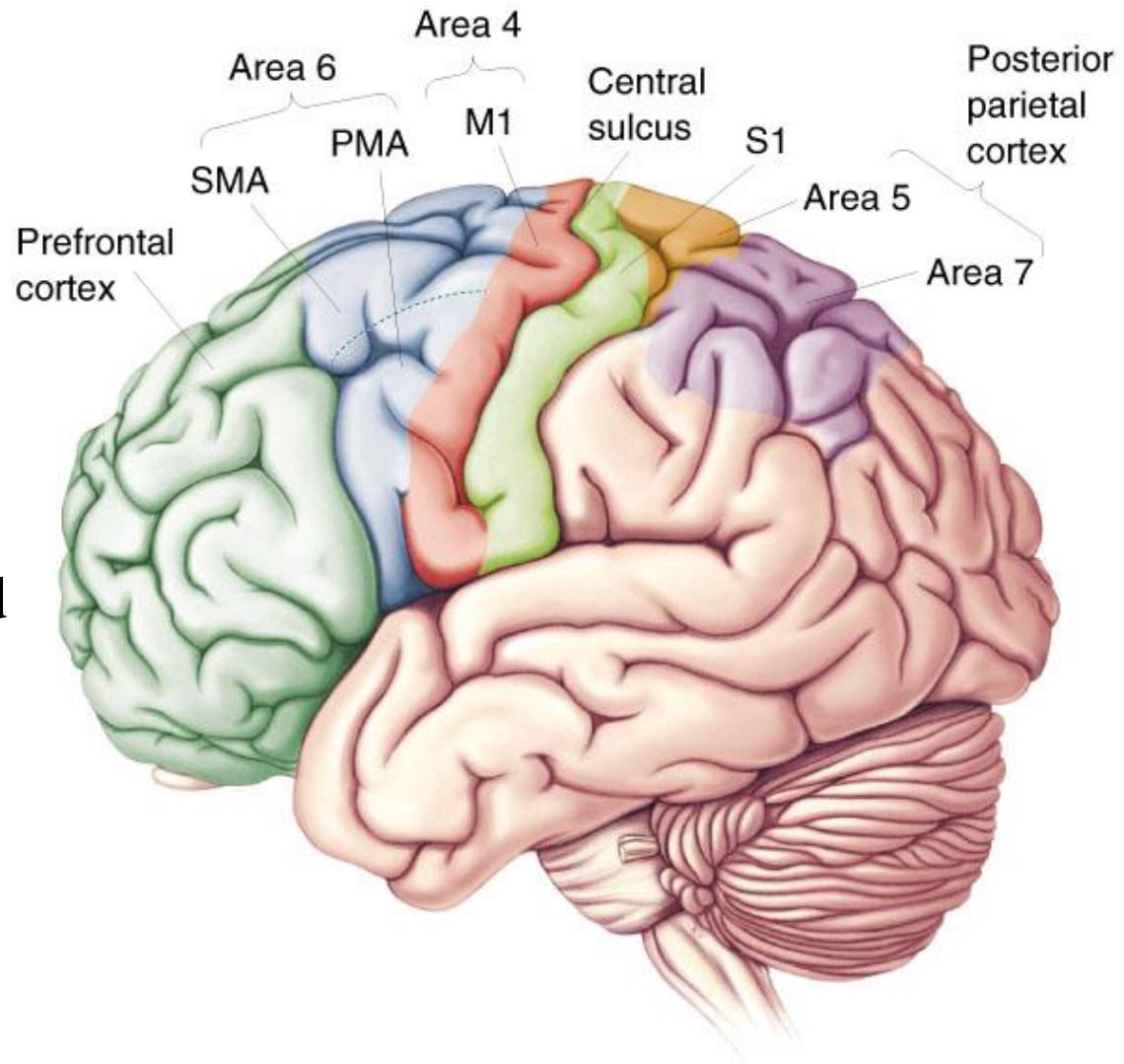
Brain areas

Figure 14.7

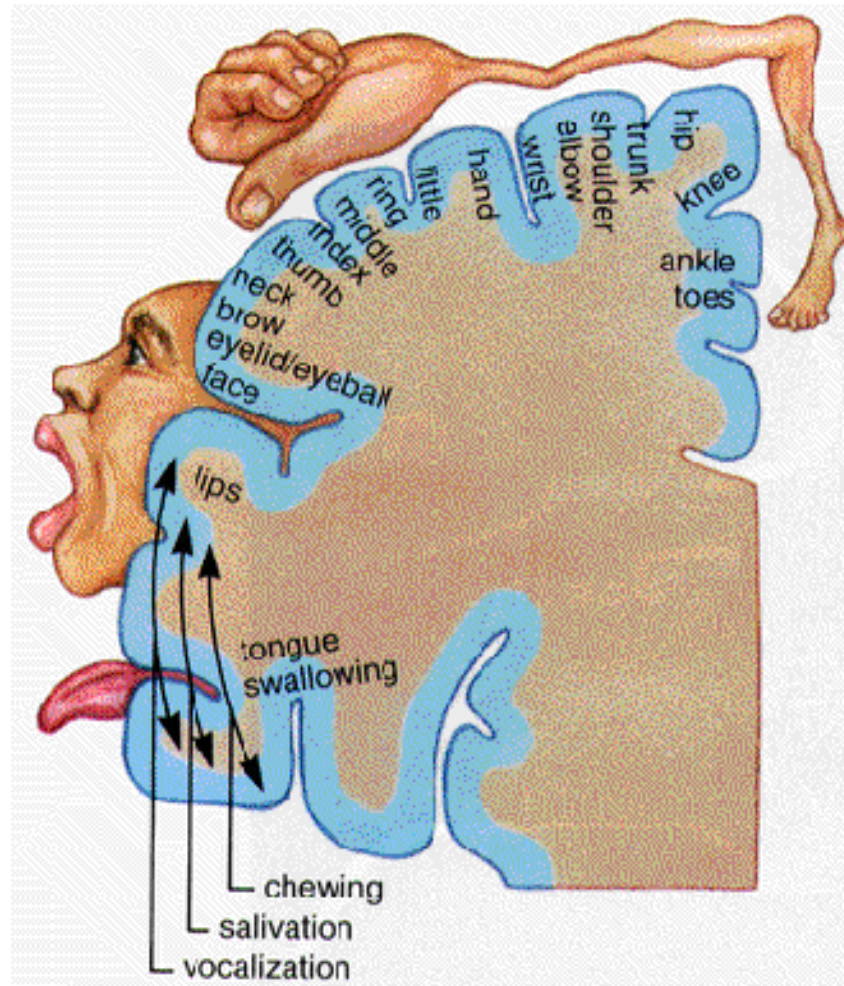
Areas of neocortex intimately involved in the planning and instruction of voluntary movement. Areas 4 and 6 constitute motor cortex.

**Human area 4 = M1 =
primary motor cortex**

**Human area 6 =
premotor area (PMA) and
supplemental motor
area (SMA) = premotor
cortex**

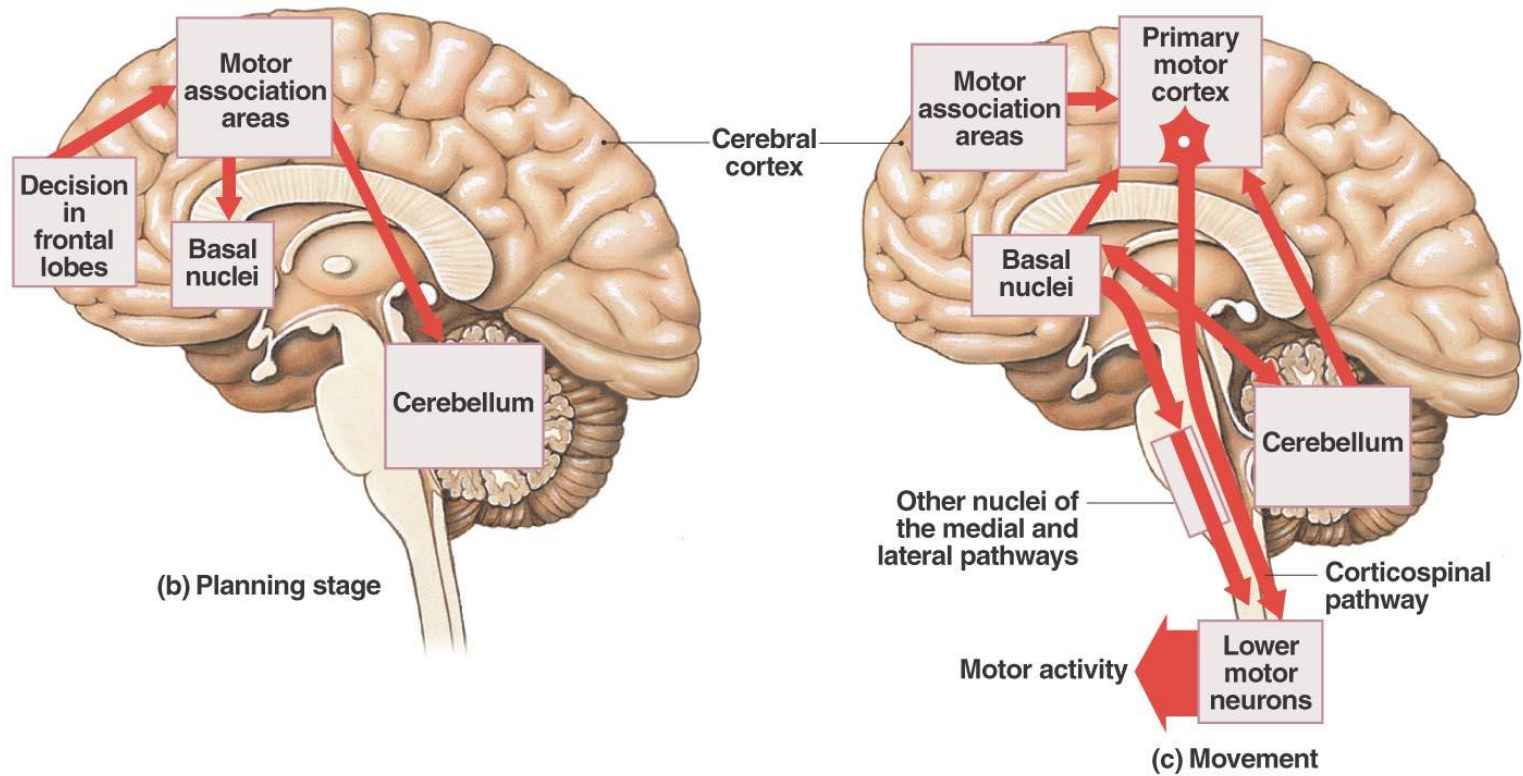


Brain areas



* This graphic representation of the regions of the Primary Motor Cortex and Primary Sensory Cortex is one example of a HOMUNCULUS (“Little man”)

Brain areas



Brain areas: Summary

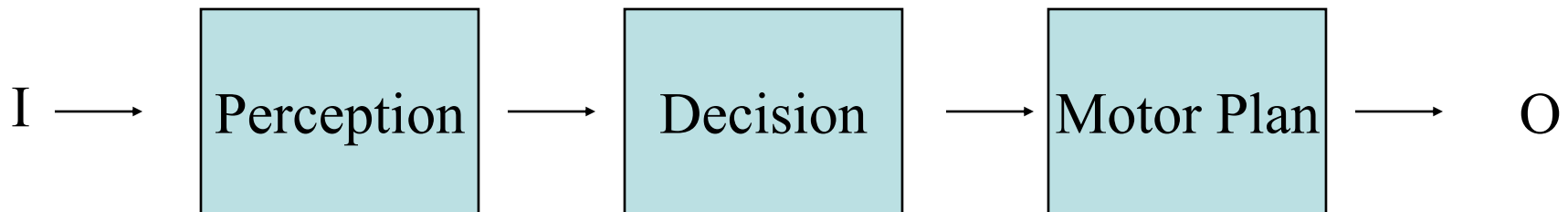
- Star Players of the (Somato-)Motor System
 - Basal Ganglia (BG), Cerebellum (CB), Motor Cortex (Primary, Secondary, Supplementary)
- Key Roles
 - Superior Colliculus (SC): control reflexes
 - BG: Voluntary control, Reinforcement learning
 - CB: Coordinates Complex motor patterns
 - Cortex: Planning and execution of voluntary movements

Motor Control vs Learning

- Motor Control: Understanding the execution of the events (physiological or behavioural / psychological description) that lead to skilled human movement
- Motor Learning: involves changes in the execution of the events (physiological or behavioural/ psychological description) that lead to **improvements** (skilled) in human movement

Motor Control

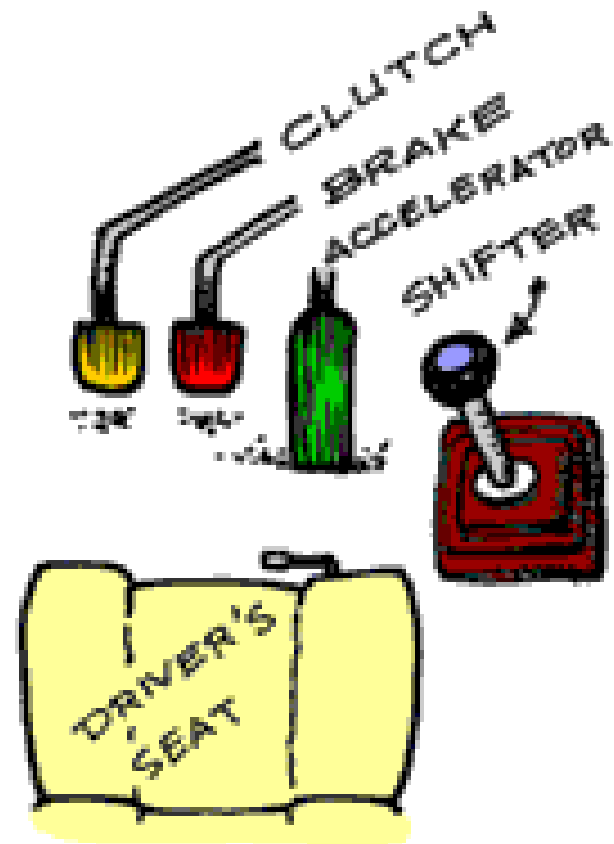
- Open-Loop (memory-driven)
 - movements are completely pre-planned and subsequent feedback during the movement doesn't change the nature of the plan
 - Ex.: over-learned movements such as signing your signature



Open

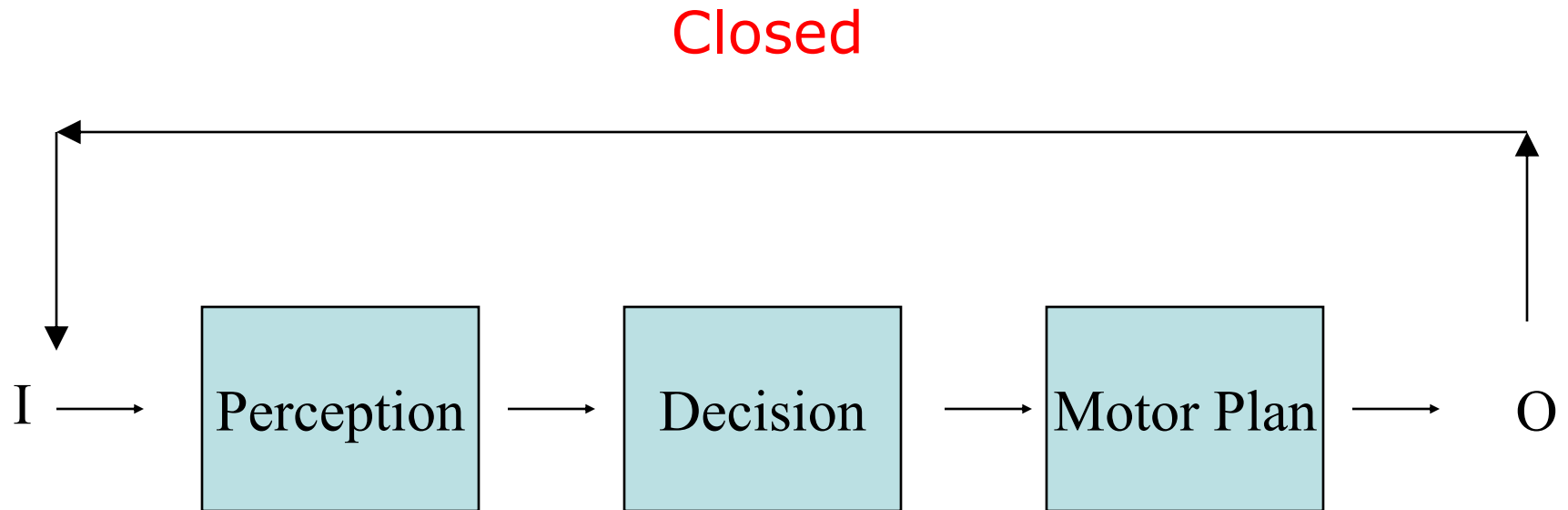
Motor Control

- Closed-Loop
 - Using feedback or closed-loop processing



Motor Control

- Closed-Loop
 - Using feedback or closed-loop processing



Assignment

- Task 1
 - Determine your handedness using Edinburgh Handedness Inventory
- Task 2
 - Write your name or signature with your non-dominant hand.
 - Do this 20 times, and see how your writing is improving.
 - Compare the first with your last trial and also with the sample when you write or sign with your dominant hand.
- Task 3
 - Write a brief note on William Calvin's "Throwing Madonna" theory for the evolution of handedness
- Task 4
 - Give four examples of usability issues versus handedness in design

Effector Dominance

- Edinburgh Handedness Inventory (Oldfield, 1971)
 - Consider each of the 10 activities listed below and indicate which hand you prefer using when engaged in each of these different tasks:
 - Writing, Drawing, Throwing, Scissors, Toothbrush, Knife (without fork), Spoon, Broom (upper hand), Striking match (match), Opening Box (lid)
 - Your Laterality Quotient = $(\text{No. of Right answers} - \text{No. of Left answers}) \times 10$.
 - +100: Extreme Rt-handed; -100: Extreme Lt-handed; 0: Ambidextrous